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Exploring the influence of moral development, religiosity, and AI self-efficacy on ethical AI use among university students in Ghana and Namibia

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Abstract
The study examined the extent to which university students' moral development, religiosity, and AI self-efficacy affect ethical use of AI. Using a cross-sectional research design, data were gathered from 1000 university students, and analysed using partial least squares structural equation modelling (PLS-SEM) and descriptive statistics. The descriptive results revealed that students possess a moderate level of moral development, religiosity, AI self-efficacy and are moderately engaged in ethical use of AI. The PLS SEM analysis, on the other hand, revealed that ethical use of AI is significantly predicted by students' moral development, religiosity, and AI self-efficacy. It was further revealed that AI self-efficacy not only influence ethical use behaviour but also serves as a significant mediating variable that connects religiosity and morality to ethical use behaviours. These results indicate that fostering ethical AI use demands a dual approach: (1) strengthening moral and religious values that underpin academic integrity, and (2) systematically building students' practical competence and confidence in using AI tools. It is therefore recommended that lecturers should exemplify integrity, incorporate moral case studies and mentor students on ethical decision-making with regard to AI.

Keywords Moral development, Religiosity, AI self-efficacy, Ethical AI use, Artificial intelligence, Ethics

1 Introduction

Artificial Intelligence (AI) has come to stay and is permeating all spheres of life: health, agriculture, music, education, 71. In the education landscape, especially in tertiary education, AI offers tremendous opportunities for innovation and efficiency. It is utilised by both lecturers and students for academic work, social and personal issues. Some of these AI tools include Grammarly, Writefull, ChatGPT, Ximilar, Cognitive scale, etc. However, there have been serious concerns about misuse or unethical use of AI by university students for their academic, social and personal lives. Scholars and stakeholders

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in Universities in Africa fear students are losing their criticality, creativity and skills due to the unethical use of AI. Unethical use of AI leads to the neglect of the principles of fairness, transparency, accountability, and respect for privacy. Literature informs that ethical behaviours are influenced by socio-cultural and psychological traits. Some of these traits are morality, religiosity and AI self-efficacy. It is imperative to understand how these traits jointly shape the ethical attitudes and behaviours of university students towards AI usage.

Generative artificial intelligence (Gen AI) technologies such as ChatGPT, Meta AI, Quillbot, Consensus, Grammarly, among others, have become valuable assistive academic technologies capable of enhancing the learning experience of students, especially those in higher education levels across the world [5–7]. Scholarly discourses on the significance of these assistive technologies have demonstrated that students primarily utilise them to enhance their overall learning experience and academic progress. It has been empirically established globally that AI provide students with significant academic opportunities such as personalized learning experience, self-assessment opportunities [32, 69, brainstorming of ideas [33, 6134], retrieval of academic information 7021], as well as the improvement of their academic writing prowess [6, 27, 39, 55]. Despite the well-documented academic benefits associated with AI use among students in various educational institutions globally, a rise in the unethical use of these technologies among students in higher education is particularly concerning [24, 28]. Several studies have also revealed that AI improves lesson preparation, customised learning, feedback, study materials, quizzes, exam questions, language learning support, virtual teaching support, tutoring, and critical thinking can all be greatly impacted by ChatGPT4961 They, however, highlight the moral dilemmas and difficulties surrounding AI application, includingdata security and privacy, openness, and conformity to educational ideals275161Here, it is genuine that using AI-generated content can lead to producing fake outputs if the user is incompetent in using the tool.

Ensuring ethical use of AI tools in higher education domains remains a serious concern for institutional policymakers globally [3]. Extant studies have demonstrated that instead of leveraging the scholarly and cognitive benefits associated with Gen-AI technologies, several students rather use these tools unethically to commit several forms of academic dishonesty [17, 45, 52, 53], a practice that questions academic integrity, education quality, and the quality of students churned out by various educational institutions in this 21st-century era. Amid these concerns, several scholars have identified the potential range of factors that account for the unethical use of AI in higher education. Surahman and Wang [52, 53], for instance, attribute unethical use of AI to students' laziness to learn, peer influence, academic workload, and ease of information access. In the African context, Onyejegbu [40] argue that the unethical use of AI among students could probably be a reflection of the lack of institutional regulatory bodies in higher education, the lack of experts, and the lack of a proper AI ethics course and AI code of conduct to regulate its use among students.

The integration of AI in education requires a structured framework to ensure its ethical use, emphasising the importance of ethical principles such as transparency and accountability [18]. AI self-efficacy, which refers to a user's confidence in their ability to effectively use AI technologies, might logically be hypothesised to influence their capacity to engage with AI in an ethically responsible manner. However, this relationship is

not explicitly explored or quantified in the studies reviewed. Additionally, the importance of fostering ethical AI literacy is emphasised in several works, suggesting an indirect relationship where higher self-efficacy in AI could enhance a student's ability to understand and apply ethical frameworks effectively [38]. This notion is supported by discussions around the requirement for robust pedagogical strategies that incorporate both AI literacy and ethics into curricula to cultivate informed and responsible users. Moreover, ethical issues such as AI-driven decision-making, student data privacy, and algorithmic bias are prominent concerns in educational contexts [54]. These challenges highlight the need for students to possess not just technical proficiency (or self-efficacy) but also a thorough understanding of ethical principles to guide their use of AI technologies responsibly. Interestingly, some discussions imply that while AI technologies offer transformative potential in education, there are still significant ethical concerns, like privacy and bias, that must be addressed to ensure responsible adoption [43]. The development of self-efficacy in AI usage could potentially empower students to engage with these technologies more responsibly, though the literature lacks explicit investigations or evidence on this potential linkage.

In this current study, ethical use of AI is conceptualised primarily within the domain of academic integrity, encompassing various behaviours like proper acknowledgement of AI-generated content, avoidance of plagiarism, not verifying content, and using AI as an assistive tool rather than a co-pilot or substitute for learning. While we acknowledge broader ethical concerns like data privacy and algorithmic bias, the operational focus here is on academic conduct. Furthermore, AI self-efficacy refers to using AI tools for academic tasks. The quest to find the possible factors influencing unethical use of AI, as well as the lasting solutions, is ongoing in the scholarly literature. Yet, a significant but often neglected factor, which is of principal interest to us in the current study, is the role of moral development in explaining ethical use of AI among university students. This is particularly important because there seems to be an inextricable nexus between morality and adherence to ethical standards, which explains why Hammerschmidt [22] highlights the role of morality in ethical AI use.

1.1 Context and justification

AI technologies have transformed how university students learn, communicate, and make decisions. University students in Africa use AI for academic work, assignments, projects, research, etc. Lecturers in some of these universities share experiences on how students use AI tools to generate responses to assignments and write research reports. Anecdotal evidence from universities in Ghana and Namibia suggests some students use AI tools at the neglect of the principles that guide its use. This brings up the issue of ethical use of AI among students. The unethical use of AI by university students does not only emanate from their lack of technological competence but also from critical attitudinal and personal traits. How do we bring a stop to the unethical use of AI among university students in Africa? Previous studies on ethical use of AI hint at the influence of morality, religiosity and AI Efficacy.

There is a dearth of literature and knowledge on how morality, religiosity, and AI self-efficacy together influence students' Unethical engagement with AI technologies. Khan [26] researched how the attitudes towards AI of religious tourists affect their AI self-efficacy and engagement in AI. Ayanwale et al. [8] studied factors that support pre-service

teachers' engagement in learning artificial intelligence. Opesemowo and Ndlovu [41] used autoethnography as the methodology and qualitative content approach to analyse data. Isiaku et al. [23] in Turkey studied the evolving role and impact of ChatGPT in higher education, trying to understand the applications, benefits and challenges associated with ChatGPT, focusing on its potential to enhance teaching, learning and assessment while addressing ethical considerations in the educational context. It has been found that among the various studies, none of them seems to have studied the theoretical development of how morality, religiosity, and AI self-efficacy together influence university students' Unethical engagement. This creates a critical gap, especially in universities in Africa and undermines efforts to foster ethical use of AI in the higher education landscape. Universities in Africa, especially the University of Cape Coast (UCC) and Namibia University of Science and Technology (NUST), will leverage the findings of this study to develop effective policies, programmes, strategies and interventions to curb the attitudinal, psychological and ethical dimensions of AI use by students. Undergirded by Deontology (Ethics of Right), the study aims to establish the influence of university students' morality, religiosity, and AI self-efficacy on their ethical use of AI. We attempt to fill the identified scholarly gap by addressing the following hypotheses:

H1: University students' religiosity will have a significant positive effect on their ethical AI use

H2: University students' moral development will have a significant positive impact on their ethical AI use

H3: University students' AI self-efficacy will have a significant positive effect on their ethical AI use

H4: The AI self-efficacy of university students will mediate the relationship between their religiosity and their ethical AI use

H5: The AI self-efficacy of university students will mediate the relationship between their moral development and their ethical AI use

2 Literature review

2.1 Theoretical foundation

The study assimilates two theories to provide robust, multi-layered explanation of ethical AI use.These theories are the Kohlberg's theory of moral development and the socio-cognitive theory of Bandura. The theory of moral provides the foundational moral reasoning that informs a students' ethical compass, while the social cognitive theory (SCT) introduce the critical mechanism of self-efficacy, which determines a student's capacity to translate their ethical intentions into actual behaviour. This theoretical triangulation justifies the use of PLS-SEM, a method well suited for testing such complex predictive models involving mediating variables.

Kohlberg's Theory of moral development (1984) is pivotal in the understanding of how individuals reason about moral dilemmas they face in everyday life. This theory explains how individuals develop morally by formulating six stages and organising them in three main levels [50]. The levels of moral development are: pre-conventional (morality is defined by obedience and self-interest), conventional (emphasises conformity to social expectations and authority), and post-conventional (reflects principled reasoning based on universal values of justice, fairness, and human dignity). This theory emphasises that moral reasoning is the basis for moral behaviour (Wang 2009). Studies [57, 60]

have shown that University students with higher moral reasoning are more inclined to act ethically. Considering the ages of university students in relation to the theory, they will be operating between conventional and post-conventional levels. These levels focus on conformity to social expectations and authority as well as universal values of justice, fairness, and human dignity. This theory clarifies how university students' moral development will shape their ethical awareness and decisions in situations of ethical use of AI. For example, students at the conventional level may avoid unethical AI use (such as unacknowledged text generation) because of institutional rules, while students at the post-conventional stage may resist such practices due to internalised values of honesty and fairness [47]. Kohlberg's theory also helps explain how religiosity can strengthen adherence to moral values, which will in turn influence their ethical use of AI.

The SCT advanced by Bandura [9, 10] proposes that human behaviour is the outcome of the interaction between personal factors, behavioural patterns and environmental influences (reciprocal determinism). The theory posits that outcome expectancies and self-efficacy play a central role in guiding and modifying the environment. Self-efficacy is therefore a central construct to this theory. Self-efficacy is an individual's belief in their ability to organise and execute actions required to achieve specific outcomes [10]. This implies that self-efficacy impacts enthusiasm, determination, tenacity, and resilience, thereby shaping how individuals behave when confronted with challenges or ethical dilemmas. Research shows that students with higher self-efficacy are more likely to engage in self-regulated learning, adopt new technologies, and act in ways consistent with their values [48, 63]. In this study, self-efficacy helps us understand university students' ability to resist pressures to engage in unethical practices related to AI use. The theory highlights the mediating role of AI self-efficacy. While moral development and religiosity may shape students' values, norms, and ethical orientations, it is self-efficacy that empowers them to translate these values into ethical practices [56]. University students who believe they can skilfully use AI tools are better positioned to avoid unethical practices such as over-reliance on unacknowledged AI-generated work or misuse of AI in ways that contravene academic integrity.

2.2 Empirical studies

2.2.1 Morality and ethical use of technology

Morality has been increasingly researched at the university level as a measure of ethical conduct. McCabe, Butterfield, and Treviño [35] discovered that students with a higher moral identity were less likely to cheat, indicating moral identity serves as a protective mechanism against unethical academic behaviours. One study explored the factors influencing nursing students' intentions to use AI and found that ethical awareness, in conjunction with digital literacy and self-efficacy, significantly affected their readiness to adopt AI technologies ethically. This indicates that moral sensitivity, which is a component of moral development, plays a role in how students perceive and use AI technologies [1, 2]. Additionally, research has shown that there is an impact of AI integration on students' educational experiences, which includes fostering critical thinking and promoting ethical practices. The incorporation of AI in education provides tailored support, which enhances students' learning experiences while promoting ethical considerations [49]. Another study illustrated the importance of integrating ethics into AI education, which suggests that students need to understand AI not only as a technical subject but

also in terms of its societal and ethical implications. Such education can potentially enhance students' moral awareness and ethical use of AI technologies [62]. Moreover, AI's role in education poses ethical challenges such as data privacy, algorithmic bias, and accountability [19]. The ethical challenges and dilemmas in using AI in education must be introduced to students to help them navigate these complexities responsibly [4]. Collectively, these findings highlight that morality, irrespective of whether it is based on identity or disengagement processes, indeed influences students' ethical decision-making, including students' relations to technologies like AI.

2.2.2 Religiosity and ethical decision making

Students' religiosity has also been found to be a significant predictor of ethics. Rettinger and Kramer [44] discovered that students who identified as more religious were less likely to cheat than students who identified as less religious. In the same vein, Johnson et al. [25] demonstrated that religiosity is associated with ethical sensitivity and the propensity to make moral choices. These results imply that religious worldviews, which supply moral discourses, would be guiding the principles for ethical reasoning of the ethical and responsible use of AI technology in university students. The question of whether there is a statistically significant relationship between university students' religiosity and their ethical AI use is intriguing, especially when considering the available literature on religiosity and ethical behaviour. One study exploring religiosity and ethics among business students suggests that religiosity significantly predicts ethical awareness in several contexts. It found that religiosity, measured through church attendance, was a predictor of students' responses in ethically charged situations [14]. This implies a potential link between how deeply students engage with their religious practices and their ethical decision-making processes. Furthermore, another study analysed the influence of ethics instruction, religiosity, and intelligence on cheating behaviours, indicating that students with higher religious engagement were less likely to cheat [11]. Although these studies focus on broader ethical behaviour rather than specifically on ethical AI use, they suggest that religiosity can influence ethical perspectives. In an exploration of how religiosity impacts emotional responses to AI, significant differences were found in emotions like fear and anger towards AI among students with varying levels of religiosity. More religious students often exhibited intense emotions towards AI technologies [29]. This could imply that religious perspectives might indirectly affect how students perceive and ethically use AI, potentially mediated through their emotional responses. The literature also discusses how religiosity can mediate ethical problem recognition and behavioural intentions. One study found that commitment to moral self-improvement, which is related to religiosity, was a significant predictor of ethical problem recognition and intentions [30]. These findings suggest that religiosity, when intertwined with a commitment to ethical self-improvement, could influence ethical decisions, possibly extending to AI usage.

2.2.3 AI self-efficacy and ethical AI use

One study highlights that the integration of AI in education requires a structured framework to ensure its ethical use, emphasising the importance of ethical principles such as transparency and accountability [18]. Venkatesh and Bala [58] also found that higher technology self-efficacy not only predicts adoption but also means that new technology

will be used less irresponsibly or unethically. These results highlight the importance of confidence in promoting responsible behaviour in AI settings. 64found that students with technical AI training were very confident in ethical applications. AI self-efficacy, which refers to a user's confidence in their ability to effectively use AI technologies, might logically be hypothesised to influence their capacity to engage with AI in an ethically responsible manner. However, this relationship is not explicitly explored or quantified in the studies reviewed. Additionally, the importance of fostering ethical AI literacy is emphasised in several works, suggesting an indirect relationship where higher self-efficacy in AI could enhance a student's ability to understand and apply ethical frameworks effectively [38]. This notion is supported by discussions around the requirement for robust pedagogical strategies that incorporate both AI literacy and ethics into curricula to cultivate informed and responsible users. Moreover, ethical issues such as AI-driven decision-making, student data privacy, and algorithmic bias are prominent concerns in educational contexts [54]. These challenges highlight the need for students to possess not just technical proficiency (or self-efficacy) but also a thorough understanding of ethical principles to guide their use of AI technologies responsibly. Interestingly, some discussions imply that while AI technologies offer transformative potential in education, there are still significant ethical concerns, like privacy and bias, that must be addressed to ensure responsible adoption [43]. The development of self-efficacy in AI usage could potentially empower students to engage with these technologies more responsibly, though the literature lacks explicit investigations or evidence on this potential linkage.

3 Methods

3.1 Research design and participants

Most relevant to this study is the positivist paradigm of social and behavioural research, characterised by a focus on objectivity, observable and measurable variables, and the testing of hypotheses [15]. Positivism aspires to study social phenomena (e.g. ethical AI use) using scientific methods just as natural sciences. The question is to find out the consequences of moral development, religiosity, and AI self-efficacy on ethical AI use. The methodology employed in gathering quantitative data from university students was the cross-sectional survey design from students in Ghana and Namibia. This perspective enables relationships between variables to be subjected to statistical analysis [46]. Undergraduate and postgraduate university students from Ghana and Namibia were involved in the study to enable us to establish contextual dynamics with the ethical use of AI. Stratified random sampling ensured representation across the country. While the sample for Namibia (n = 137) is smaller than Ghana's (n = 863), this reflects the relative sizes of the population in the accessible universities. Stratified random sampling was employed within each country to ensure representation across academic disciplines (Table 1).

Before data collection, authorisation was requested from the University of Cape Coast's Department of Arts Education. The department also provided an introductory letter to facilitate access for students in different universities. Regarding their participation, participants' informed consent was requested. Respondents who did not want to answer questions were not forced to do so. Since no identification was needed from participants, confidentiality and anonymity were guaranteed.

Table 1 Demographic characteristics of pre-service teachers (n = 1000)

Variable	Subscale	No	%
Sex	Male	403	40.3
	Female	597	59.7
Level of education	Level 100	110	11.0
	Level 200	252	25.2
	Level 300	160	16.0
	Level 400	307	30.7
	Master's students	171	17.1
Country	Ghana	863	86.3
	Namibia	137	13.7

3.2 Instrumentation

The questionnaire was the instrument designed to collect data on the influence of university students' morality, religiosity, and AI self-efficacy on their ethical use of AI. The scale was pilot-tested with a sample of 40 participants, yielding high reliability scores for all the variables: 0.981 and morality (α = 0.991), religiosity (α = 0.983), AI self-efficacy (α = 0.973) and ethical use of AI (α = 0.835). The composite reliability (CR) for the model was 0.955, which is excellent. The question was divided into four categories or sections, labelled A, B, C, and D. Section A covered the background information of the respondents (Student characteristics). Section B looked at 15 adapted items from Mensah and Kyei [36] on moral development, Section C looked at 20 adapted items from Mensah et al. [37] on religiosity, Section D, containing 12 items on AI self-efficacy, was self-developed, and Section E collected data on the ethical use of AI with 15 items**.** Responses were captured using a five-point Likert scale: Strongly Agree (5), Agree (4), Uncertain (3), Disagree (2) and Strongly Disagree (1). The 12-item AI self-efficacy scale was self-developed based on Bandura's guidelines for constructing self-efficacy scales. Items were stated to reflect confidence in performing AI-related academic tasks, eg. I am confident I can use AI to brainstorm ideas for my assignment. Content validity was established through review by three experts in educational technology. The pilot study showed excellent reliability (α = 0.973), which was confirmed in the main study.

3.3 Data processing and analysis

For data analysis, both descriptive and inferential statistical methods were employed. The survey responses were coded and numbered to ease identification, and the data were thoroughly reviewed for completeness and accuracy. Any inconsistencies were addressed without altering the original responses. Descriptive statistics (frequencies, percentages, means, and standard deviations) were used to interpret responses to the first and second research questions. Data was analysed with both descriptive and inferential statistical measures. Statistical software, such as SPSS v29 and SmartPLS SEM, was used. Before conducting these analyses, the assumptions of normality and other relevant statistical assumptions were checked and satisfied.

3.3.1 Ethical consideration

The present study was conducted in accordance with rigorous ethical standards to protect the participants and maintain scientific integrity. First, the institutional review board of the University of Cape Coast gave a waiver as a requirement for ethical approval, given the low risk of the study. Further, the first page of the online form outlined the

study's purpose, benefits and rights to withdraw. Participants provided consent by clicking "I agree to participate' before proceeding to the questionnaire. Anonymity was protected through the elimination of personally identifying information, and information was secured in password-protected files. The research followed the Belmont Report's Principles of Respect, Beneficence, and Justice [13], educational research ethics. Sensitive issues were raised in neutral, culturally appropriate language, and findings were presented truthfully.

4 Results

4.1 Descriptive results

Presented in Table 2 are the descriptive results on students' perceived level of moral development, religiosity, AI self-efficacy and their ethical use of AI in performing their academic tasks. The results were analysed using means and standard deviation. The mean and standard deviation scores were interpreted as follows: 1.00–1.9 = (low), 2.0–3.4 = (moderate) and 3.5–5.0 = (high).

The results, as presented in the table, suggest that university students collectively demonstrate a moderate level of moral development (M = 3.1; SD = 1.5). Nonetheless, the relatively higher SD suggest heterogeneity in the students' rating of their moral development. This means that several students are highly moral, and others have a very low level of moral development. Similarly, the results demonstrated that the students are moderately religious (M = 3.1; SD = 1.5). With high standard deviation values, there is a high chance that many students are either highly religious or report an extremely low level of religiosity. This demonstrates a lack of consensus on the level of religiosity among the respondents in the current study. The AI self-efficacy of the respondents was also moderate (M = 3.0; SD = 1.4). The corresponding SD demonstrates heterogeneous responses. This means that there are a few extreme responses with regard to their level of AI self-efficacy. Finally, it was discovered that the use of AI among university students engaged in this study is moderately ethical. The large standard deviation suggests that there are several students who may be engaging in unethical use of AI in performing their academic tasks.

4.2 Hypothesis

4.2.1 Measurement model assessment

The measurement model was evaluated to confirm the quality of the model. This included an assessment of the reliability, convergent validity, and discriminant validity.

4.2.1.1 Reliability and convergent validity According to scholars , the first indicator of internal consistency is the indicator loading, which demonstrates a good internal consistency in the measurement when it obtains a loading of 0.7 or better. In different cases, however, loadings below 0.7, especially those that load as low as 0.4, are retained in the

Table 2 Descriptives

Construct	M	SD
Students' moral development	3.1	1.5
Students' religiosity	3.1	1.5
University AI self-efficacy	3.0	1.4
Ethical use of AI	3.2	1.3

measurement when they do not affect other measurement quality criteria, especially the average variance extracted AVE. Based on this, we can confirm that the measurement has good internal consistency because most of the loadings are above the 0.7 threshold, and those that loaded below 0.7 were retained because they did not affect other quality criteria in the model. Another measure of internal consistency is Cronbach's alpha and the composite reliability. It is recommended that the Cronbach's alpha and the composite reliability values must be 0.7 or above. As presented in Table 3, all the values for Cronbach's alpha and the composite reliability are above the recommended threshold, confirming that the measurements are reliable.

After the internal consistency assessment, the validity of the measurement model was also assessed. This was done using the average variance extracted. The convergent validity was used to ascertain how the items are reflective of their underlying construct. According to scholars65 AVE values of 0.5 and above are an indication that a group of items are indeed measuring a unique construct. The results, presented in Table 3 and depicted in Fig. 1, demonstrate that there is strong convergent validity in the current model because all the constructs achieved AVE values above the 0.5 threshold.

4.2.1.2 Discriminant validity The discriminant validity of the model was evaluated to ascertain how different the variables are from each other. In other words, it was used to find out the conceptual distinctiveness among the various constructs in the model.

Table 3 Convergent validity and reliability results

	Loadings	α	CR (rho_a)	CR(rho_c)	AVE
AI-EFF		0.948	0.951	0.958	0.763
AISE1	0.872				
AISE12	0.878				
AISE3	0.848				
AISE4	0.860				
AISE6	0.855				
AISE8	0.897				
AISE9	0.904				
ETH		0.945	0.957	0.955	0.753
EUoAI10	0.861				
EUoAI11	0.819				
EUoAI12	0.919				
EUoAI13	0.893				
EUoAI14	0.889				
EUoAI15	0.899				
EUoAI9	0.785				
MD		0.950	0.950	0.964	0.870
Q1	0.928				
Q15	0.933				
Q5	0.930				
Q9	0.940				
REL		0.943	0.945	0.954	0.746
R10	0.884				
R2	0.805				
R20	0.836				
R6	0.896				
R7	0.900				
R8	0.838				
R9	0.884				

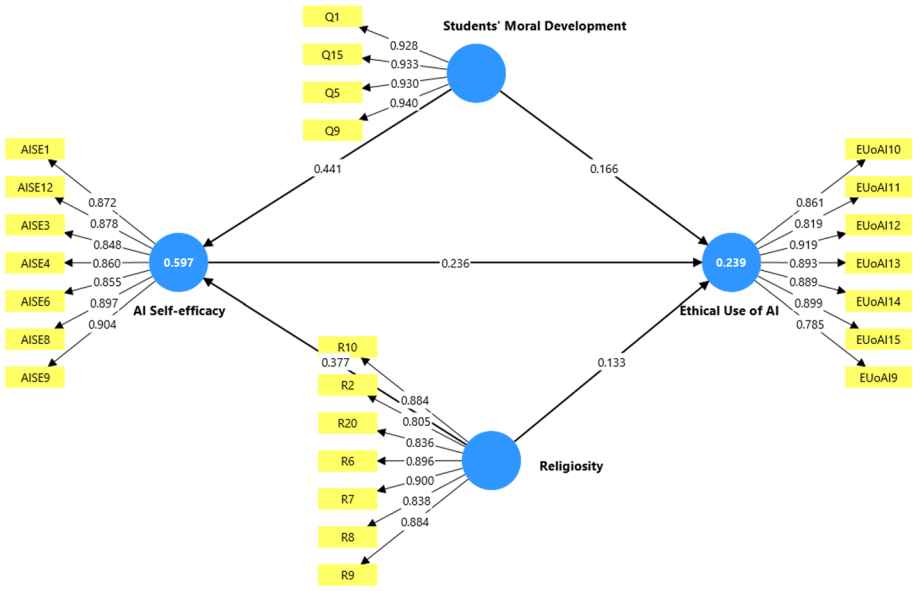

Fig. 1 PLS-SEM measurement model

Table 4 Discriminant validity results

HTMT	AI-EFF	ETH	MD	REL
AI-EFF				
ETH	0.471			
MD	0.770	0.458		
REL	0.757	0.449	0.822	

Fornell–Larcker	AI-EFF	ETH	MD	REL
AI-EFF	0.874			
ETH	0.454	0.868		
MD	0.736	0.444	0.933	
REL	0.722	0.433	0.781	0.864

Scholars recommend the use of the Heterotrait-Monotrait (HTMT) ratio and the Fornell-Larcker criterion to determine discriminant validity in structural equation modelling analysis . The results of the discriminant validity assessment are presented in Table 4.

The HTMT ratio determines discriminant validity by comparing the correlation of items of the same construct with the correlation of items across constructs in a model. To ensure discriminant validity among the constructs, scholars recommend that the HTMT values must be less than 0.8 or 0.9 in extreme cases. The HTMT ratio between moral development and religiosity was 0.822, slightly above the conservative threshold of 0.80, but the more liberal 0.90. This suggests that while these constructs are conceptually associated, they are sufficiently distinct for analysis in this exploratory model. Since the HTMT values of the current model are below the recommended threshold, it is safe to declare that the current model has discriminant validity. The Fornell-Larcker criterion also confirmed the discriminant validity of the current model, as the square root of the AVE of each of the constructs in the model is significantly higher than the average of correlations among the constructs, as 65and 66recommend.

4.2.1.3 Common method bias (CMB) Issues of CMB arise when the same method is used for gathering data on both criterion and predictor variables in a particular study.

Table 5 CMB results

	AI-EFF	ETH	MD	REL
AI-EFF		2.481		
ETH				
MD	2.565	3.049		
REL	2.565	2.918		

Table 6 PLS-SEM Bootstrapping results

	Paths	B	SD	T	*p*	F^2	R^2	Q^2	Decision
H1	REL -> ETH	0.133	0.068	1.970	0.049	0.008	0.239	0.208	Accept
H2	MD -> ETH	0.166	0.077	2.164	0.031	0.012			Accept
H3	AI.EFF -> ETH	0.236	0.067	3.524	0.000	0.029			Accept

This can always lead to undesirable response patterns, which culminate in inflated variances among the constructs in the model. It is, therefore, important for researchers to evaluate the existence of CMB to provide authenticity to regression results. There are several ways of determining the existence of CMB in a particular model; however, scholars usually recommend the use of multicollinearity assessment.

CMB was assessed using both procedural and statistical remedies. Procedurally, the respondent's anonymity was guaranteed. Statistically, Harman's single-factor test was conducted, revealing that the single factor explained 38.2% of the variance, well below the 50% threshold, indicating that CMB is not a critical issue. Again, the collinearity for all constructs was below 3.3, providing additional evidence that CMB is not significantly biasing the results. Results of the multicollinearity assessment are presented in Table 5.

In assessing collinearity, the variance inflation factor (VIF) is used to ascertain the prevalence of inflated variances in the model. A model is said to be devoid of collinearity issues when the VIF of the predictors are below 3.3 or 5.0 in extreme cases66 The results for the current study suggest that there is no inflated variance among the constructs, because none of them surpass the recommended threshold. This presupposes that the results of the regression are highly reliable.

4.2.2 Structural model assessment

The principal goal of the study was to explore the predictive impact of university students' religiosity and moral development on their ethical use of AI, and also look at the mediating role of AI self-efficacy in these predictive relationships. To achieve this, the PLS-SEM bootstrap method with 5000 subsamples was used. The results of the hypotheses are presented in Table 6 and also depicted in Fig. 2.

The results presented in the table and the figure confirmed all the hypotheses set to guide the study. The first hypothesis (H1), which set out to measure the influence of religiosity on the ethical use of artificial intelligence among university students based in Ghana and Namibia was statistically significant ($\beta = 0.133$, $t = 1.970$, $p = 0.049$, $f^2 = 0.008$), although the effect size, which concerns the statistical impact of the predictor on the criterion variable, is minimal when judge against the threshold proposed by [12]. The practical implication of this confirmed hypothesis is that students who report high levels of religiosity are more likely to engage in ethical AI practices compared to those with low levels of religiosity at the university.

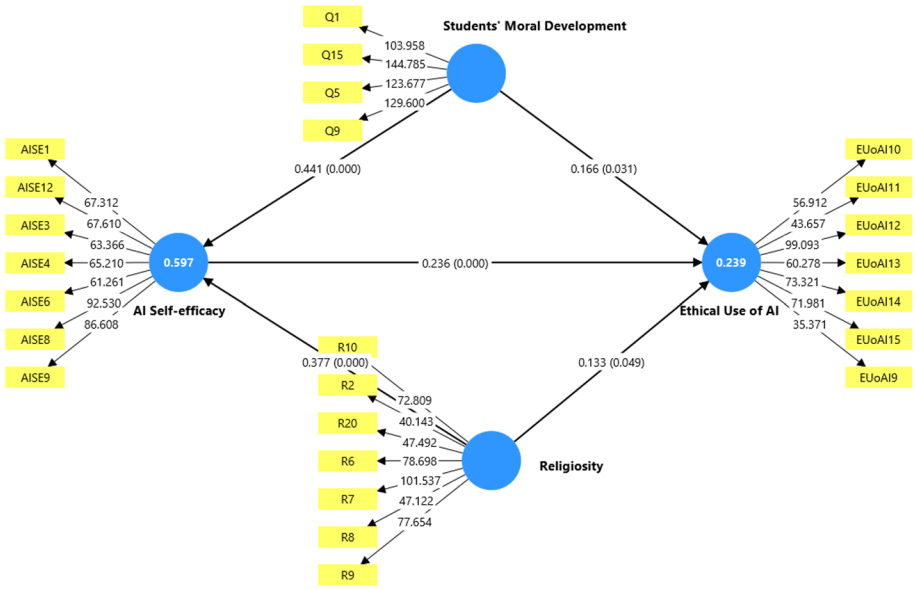

Fig. 2 PLS-SEM bootstrap results

Table 7 Mediation analysis

	Hypothesized path	Direct effect	*p*-value	Indirect effect β	95% bootstrap CI	Total effect	Mediation type	Decision
H4	MD -> AI-EFF -> ETH	0.166	0.031	0.089	[0.035, 0.148]	0.255	Partial	Accept
H5	REL -> AI-EFF -> ETH	0.133 ()	0.049	0.104	[0.038, 0.169]	0.237	Partial	Accept

The second research hypothesis (H2) investigated the influence of students' moral development on their ethical use of artificial intelligence. The results for this hypothesis were confirmed. It was found that moral development is a significant positive predictor of ethical use of AI among university students. This result presupposes that students with higher levels of moral development are likely to use AI ethically in their academic activities ($\beta = 0.166$, $t = 2.164$, $p = 0.031$, $f^2 = 0.012$). The practical implication of this result is that students with low moral development are more likely to engage in unethical use of artificial intelligence for their academic activities. This may include high levels of plagiarism, improper citation of references and the tendency to use artificial intelligence to ride their assignments without proper acknowledgement.

The study's findings also suggested that the ethical use of AI among universities is significantly contingent on their AI self-efficacy, thereby confirming the third Research Hypothesis ($\beta = 0.236$, $t = 3.524$, $p < 0.001$, $f^2 = 0.029$). The positive beta value obtained in this hypothesised relationship suggests that students who feel more capable and confident in using AI are more likely to utilise AI in an ethically sound manner. An alternative explanation for this result is that students may use AI in an unethical manner, having an unlimited sense of efficacy regarding the use of artificial intelligence in academic activities.

The mediation analysis in Table 7 confirms H4 and H5. The significance of indirect effects shows that AI self-efficacy is a partial mediator. The mediating role of AI self-efficacy in the relationship between morality and students' religiosity was found to be statistically significant, hence confirming the H4 ($\beta = 0.089$, $t = 0.166$, $p = 0.031$) and H5 ($\beta = 0.104$, $t = 0.133$, $p = 0.049$). This means moral development and religiosity influence

ethical AI use, and also exert a significant portion of the influence by boosting students' confidence in using AI, which in turn promotes ethical behaviour. The practical implication of the mediation result is that students' ethical behaviours regarding the use of artificial intelligence in academic activities are not solely dependent on their moral and religious values, but also depend on their confidence in their ability to use artificial intelligence tools effectively in performing specific academic tasks. This implies that a student might be morally upright or might be very religious in nature, but may still engage in unethical use of artificial intelligence tools if they lack the competence and confidence related to effective ways of using artificial intelligence tools for academic activities.

Finally, the coefficient of determination and predictive relevance were evaluated to confirm the explanatory power of the current model. The coefficient of determination (R^2) shows that the proportion of variance in the criterion variable explained by the Predictors is 0.239, which shows that of all the behavioural variables that could influence ethical artificial intelligence use among students, morality, religiosity, and AI self-efficacy explain 23.9% of the proportion of variance in ethical use of AI. This, according to the threshold laid down by Hair et al. (2019), demonstrates that the predictors explain the minimal proportion of the variability in the criterion variable. The predictive relevance, on the other hand, demonstrated that the predictors offered an accurate prediction of the criterion variable since the Q^2 value is more than 0, which is the recommended threshold for determining the accuracy of statistical predictions in PLS-SEM modelling (Hair et al., 2019; Sarstedt et al. 2023).

5 Discussion

This study has investigated the predictive influence of university students' religiosity, morality, and self-efficacy on the ethical use of artificial intelligence in performing academic activities. The findings of the current study confirmed the compelling role of religious and moral dimensions of individuals' behaviours in explaining their ethical behaviours in performing a given task. The study has shown that the ethical application of educational artificial intelligence tools among university students could be explained by their level of religiosity. The results of the current study underscore the fact that high levels of religiosity among students strongly predict ethical use of AI tools. This revelation suggests university students situated in the study's context, i.e., Ghana and Namibia, who strictly adhere to the ethical principles of their faith, demonstrate a lesser tendency towards using AI in an unethical way, probably because such an unethical engagement is unacceptable. The results are not surprising because they confirm existing literature on how individuals rely on their belief systems and religious faith to act ethically in various circumstances. Although a limited attempt has been made to find out how religiosity influences ethical technology use behaviours, the findings of the current study align with studies in other fields of study. For instance, aligns with the revelation that intrinsic religiousness positively influences ethical consumer behaviours [42, 59]. The findings contradict those of Kozak and Fel [29], who found that highly religious university students demonstrate a negative emotional response to AI tools and thus demonstrate a lesser likelihood of using AI.

Moral development was also found to be a statistically significant predictor of ethical use of AI among universities in Ghana and Namibia. This revelation, like that of the theoretical connection between religiosity and ethical AI use, implies that the sense of moral

uprightness among university students in Ghana and Namibia contributes immensely to their AI use behaviours. This presupposes that students who value moral standards are less likely to use AI in an unethical manner, such as using it to cheat in their assignments and other academic-related activities. The findings of the current study fall in line with previous studies that have demonstrated that morality mitigates the urge to engage in academic dishonesty and cheating among students [16, 20]. It can be concluded that morality always serves as an ethical reminder and a driver of responsible use of AI in academic-related activities.

Confidence and competence in one's ability to use AI responsibly are a crucial matter, which therefore necessitated the need for an investigation into how AI self-efficacy plays a role. The results show that students' sense of confidence in relation to the use of AI predicts their ethical use behaviours. This particular revelation was not counterintuitive because, without the required level of knowledge on how to use AI in education, one can knowingly or unknowingly engage in unethical use behaviours. It is thus not surprising that the AI self-efficacy had a significant positive predictive influence on the ethical use of AI. The flip side of the argument is that students with less AI self-efficacy may engage in unethical use behaviours. This particular finding reiterates the extant research findings on how AI self-efficacy is theoretically connected to use behaviours and behavioural intentions [1, 2, 31]. Further, it has been emphasised in this study that AI self-efficacy not only influences students' ethical AI use behaviours but also serves as a significant mediating variable between the predictors (religiosity and morality) and the ethical use of AI.

It is important to contextualise the strength of these relationships. While religiosity and moral development were significant predictors, their individual effect sizes were small. This indicates that, although they are important pieces of the puzzle, their isolated impact is limited. Moreover, the model explained 23.9% of the variance in ethical AI use. This R^2 value is common in behavioural research and highlights that other important determinants, such as institutional policies, peer culture, programme demands and specific digital literacy, are also at play and should be investigated in future research. In more sophisticated religious communities like Ghana and Namibia, spiritual beliefs can be a powerful internal instigator of conduct, potentially compensating for underdeveloped institutional AI policies. The crucial mediating role of AI self-efficacy suggests a competence-confidence bridge. A student may hold strong moral values, but without the confidence and competence with the skill to use AI effectively, they may still inadvertently engage in unethical practices like overdependence or poor citation habits.

6 Implications and recommendations

The findings of the current study emphasise the need for higher education institutions in Africa to find ways of incorporating religio-moral values and ethical principles in digital literacy development and AI educational policies. Religiosity and morality were found to be compelling determinants of ethical adoption of artificial intelligence among university students across these two countries. Rather than standalone policies, integrate AI ethics modules into existing religious and moral education courses. Frame ethical use as an extension of honesty, integrity and respect. Concerned stakeholders must design ethics-oriented curricula and workshops that reiterate responsible use of artificial intelligence, especially in the field of Education. There is also a need for instructors to emphasise the

place of morality and religiosity in enhancing ethical use of artificial intelligence, especially in a highly religious continent like Africa.

Universities should move beyond theoretical warnings. Workshops must build practical AI self-efficacy by teaching students how to use AI for brainstorming, editing and research while properly citing and acknowledging its assistance. This is, particularly, important because although morality and religiosity influence the ethical use behaviours of students, building their capacity and knowledge in the use of artificial intelligence. This can lead to the avoidance of unethical use behaviours, which may stem from ignorance of the responsible ways of using these assistive technologies.

Based on the findings, universities will confidently empower lecturers by providing faculties with training and resources to design assessments that leverage AI as a collaborative tool, reducing the temptation for academic dishonesty and fostering the very self-efficacy this study identifies as critical.

6.1 Limitations and further studies

The study's use of a cross-sectional design prevents causal inference. Second, the use of self-report measures is prone to social desirability bias, particularly for sensitive topics like morality and ethical behaviour. The results' applicability throughout Africa is restricted by the sample imbalance and the geographic concentration on just two nations. The quantitative nature of the study did not offer the opportunity to explore students' lived experiences and subjective views regarding the ethical use of AI in their academic activities. Based on this shortfall, we recommend that subsequent research studies should employ a qualitative approach to understand the subjective experiences of students regarding the ethical use of AI. Moreover, this study examines the ethics of AI use in education from a moral and religious theoretical perspective. Future research should utilise experimental or longitudinal designs, include objective measures of AI usage, verify a more thorough ethical AI use scale, and use frameworks such as UTAUT to examine the impact of peer pressure and institutional policies.

Author contributions
E.M., T.A., E.N., I.O.A., and F.O.S are jointly responsible for the conceptualisation, methodology, analysis, and report writing.

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Data availability
Data is available and will be provided on request.

Declarations

Ethics approval and consent to participate
The University of Cape Coast's Institutional Review Board gave the waiver for the requirement of ethical approval. All participants gave their informed consent to participate in the study before being able to fill in the questionnaire.

Consent for publication
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Competing interests
The authors declare no competing interests.

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